

ANDREW STEVENS

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Website Portfolio: www.andrewstevens.engineering

EDUCATION

Virginia Tech – Blacksburg, VA

B.S. Electrical Engineering

Expected: May 2029

GPA: 4.0

PROFESSIONAL EXPERIENCE

Hitachi Energy – Raleigh, NC

May 2026 - August 2026

Internship - Software Engineer

- Scoped and planned an autonomous Python report generator, defining requirements and milestones
- Built the application to parse transformer service test files and generate formatted reports, reducing documentation time from 4–6 hours to approximately 5 minutes (~98% reduction)

Andy's 3D Workshop – Ashburn, VA

March 2021 - June 2026

Owner and Founder

- Established the company and developed business plans for optimized sales growth
 - Designed and prototyped all products, including 1 patented with US Patent & Trademark Office
 - Sold products to businesses and partnered with 2 local stores in Northern Virginia
 - Created the company's online store on Etsy with over 1,600 sales and a 5-star rating
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ADDITIONAL EXPERIENCE

Hybrid Electric Vehicle Team – Virginia Tech

January 2026 - Present

Connected & Autonomous Vehicles (CAV) Team Member

- Designed Cooperative Adaptive Cruise Control in simulink to increase energy efficiency of Cruise Control by 60.3%
- Improved existing ROS2 in Python Cruise Control to account for the changed Cooperative Adaptive Cruise Control
- Training in high-voltage safety systems, powertrain architecture, vehicle sensing, control integration, and cross functional engineering processes for an electric vehicle that is part of the EcoCar Competition
- Designed and implemented intersection crossing logic in Simulink and control blocks to govern vehicle decision-making

Competitive Robotics Organization (CRO) – Virginia Tech

Canopy ECE Lead

August 2026 - Present

- Lead electrical and computer engineering architecture for an autonomous greenhouse robotic system, defining system structure, interfaces, and technical direction
- Recruited and interviewed prospective engineers, onboarded new members, and communicated project goals, system architecture, and technical requirements

Workcell ECE Member

September 2025 - September 2026

- Designed calibration and position script in Python for the system gantry that achieved precision of $\pm 0.05\text{mm}$
- Designed custom PCBs for daisy chaining limit switches for plate storage detection, eliminating 59 loose wires
- Designing a printer UI and control system to interface with the printers, providing status feedback and parameter control

Baja SAE – Virginia Tech

September 2025 - March 2026

Testing & Electronics

- Developed a custom microcontroller GPS tracker for position and velocity, to improve race performance analysis
 - Made a Python interface to visualize sensor data at any point during a race, improving diagnostics and evaluation
 - Developed telemetry for real time monitoring of vehicle health in competitions, improving reliability and safety
 - Designed custom professional PCBs in KiCAD, improving system reliability for the Baja electronics suite
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SKILLS

- Technical Skills: PCB Design, Embedded Systems, Digital Systems, Circuit Design, C++, Python, C#, MATLAB, Simulink, ROS2, Git, Microcontrollers, OpenCV